

MINOR PROJECT 5

Ollama + Streamlit – AI Study Assistant

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Batch / Section	August/A
Model	Llama 3.2
Environment	VS Code / Python / Ollama / Streamlit

1. Objective

To build a simple AI question-answering application using Ollama as a local AI model and Streamlit as the user interface, allowing users to enter prompts and receive AI-generated responses.

2. Project Overview

The AI Study Assistant is a simple web application created with Python, Streamlit and Ollama. The user enters a question in the Streamlit interface. The question is sent to the locally running Llama 3.2 model through Ollama, and the generated response is displayed on the same webpage.

3. Tools and Technologies

Python: Application programming language. **Ollama:** Runs and manages the local Llama 3.2 model. **Streamlit:** Creates the interactive web interface. **VS Code:** Development environment.

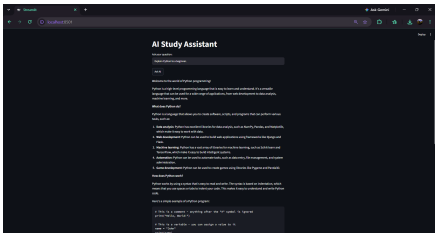
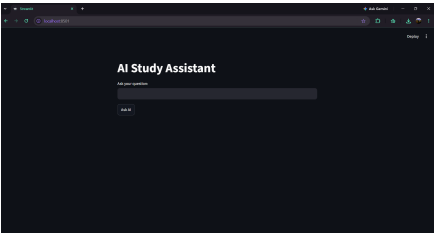
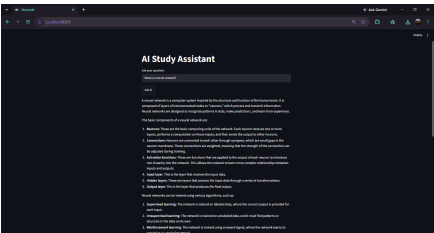
4. Implementation

The application imports the Streamlit and Ollama libraries, creates a title and text-input box, and provides an **Ask AI** button. When a question is entered, `ollama.chat()` sends the prompt to Llama 3.2. The returned message content is then displayed using Streamlit.

5. Test Prompts

1) What is artificial intelligence? 2) Explain machine learning in simple words. 3) What is a neural network? 4) Explain Python to a beginner. 5) What is the difference between AI and machine learning?

6. Screenshots

		
Main Interface	AI Response	Neural Network Response

7. Key Learnings

Learned how to run a local AI model using Ollama, connect Python to the model, build a simple Streamlit interface, send user prompts, and display AI-generated responses.

8. Conclusion

The project successfully demonstrates a basic local AI application using Ollama and Streamlit. Llama 3.2 generates responses to user questions, while Streamlit provides the interface for interaction.